

Problem Statement

Evaluate the following limit from the image "dailyintegral-limit-limit-1x1-2026-07-23.jpg":

$$\lim_{n \rightarrow \infty} \left(n + \frac{1}{2} \right) \int_0^1 x^n e^x dx$$

Detailed Solution

Let the integral be denoted as $I_n = \int_0^1 x^n e^x dx$. To evaluate the limit, we will first find a relationship between I_n and I_{n+1} using integration by parts, and then use the Squeeze Theorem to evaluate the asymptotic behavior.

Step 1: Integration by Parts

We apply integration by parts to I_n . Let $u = e^x$ and $dv = x^n dx$. Then, $du = e^x dx$ and $v = \frac{x^{n+1}}{n+1}$.

Using the formula $\int u dv = uv - \int v du$:

$$I_n = \left[\frac{x^{n+1} e^x}{n+1} \right]_0^1 - \int_0^1 \frac{x^{n+1}}{n+1} e^x dx$$

Evaluating the boundary terms gives:

$$I_n = \frac{e}{n+1} - \frac{1}{n+1} \int_0^1 x^{n+1} e^x dx$$

Notice that the remaining integral is exactly I_{n+1} . So we have:

$$I_n = \frac{e}{n+1} - \frac{1}{n+1} I_{n+1}$$

Multiplying both sides by $(n+1)$, we get a clean exact relationship:

$$(n+1)I_n = e - I_{n+1}$$

Step 2: Bound the Integral I_{n+1}

Next, we determine the limit of I_{n+1} as $n \rightarrow \infty$. For $x \in [0, 1]$, the function e^x is strictly increasing, so $1 \leq e^x \leq e$. We can use this to bound I_{n+1} :

$$\int_0^1 x^{n+1} (1) dx \leq \int_0^1 x^{n+1} e^x dx \leq \int_0^1 x^{n+1} (e) dx$$

Evaluating the outer integrals yields:

$$\frac{1}{n+2} \leq I_{n+1} \leq \frac{e}{n+2}$$

By the Squeeze Theorem, as $n \rightarrow \infty$, both $\frac{1}{n+2} \rightarrow 0$ and $\frac{e}{n+2} \rightarrow 0$. Therefore:

$$\lim_{n \rightarrow \infty} I_{n+1} = 0$$

Step 3: Evaluate the Target Limit

Using the result from Step 2 in our exact relationship from Step 1, we can find the limit of $(n+1)I_n$:

$$\lim_{n \rightarrow \infty} (n+1)I_n = \lim_{n \rightarrow \infty} (e - I_{n+1}) = e - 0 = e$$

The original problem asks for the limit of $(n + \frac{1}{2}) I_n$. We can manipulate this expression algebraically to utilize the limit we just found:

$$\lim_{n \rightarrow \infty} \left(n + \frac{1}{2} \right) I_n = \lim_{n \rightarrow \infty} \left[\frac{n + 1/2}{n + 1} \cdot (n + 1) I_n \right]$$

Because the limit of a product is the product of the limits (provided both exist and are finite), we evaluate them separately:

$$\lim_{n \rightarrow \infty} \frac{n + 1/2}{n + 1} = \lim_{n \rightarrow \infty} \frac{1 + \frac{1}{2n}}{1 + \frac{1}{n}} = 1$$

$$\lim_{n \rightarrow \infty} (n + 1) I_n = e$$

Multiplying these together yields the final result:

$$1 \cdot e = e$$

Final Answer:

$$\mathbf{e}$$